Dis-Chem Group

Omni-Channel Inventory Orchestrator

Executive Strategy & Technical Architecture Document

Proof of Concept: Not Real Data | No Official Affiliation

Disclaimer & Proof of Concept (PoC)

Status

This presentation is a Proof of Concept (PoC) only, designed to demonstrate strategic architectural capabilities.

Data Integrity

All data, figures, and projections contained herein are illustrative and do not represent real Dis-Chem operational data.

Affiliation

There is no formal connection, partnership, or endorsement between the author of this document and Dis-Chem Group.

Purpose

This document is intended for strategic architectural demonstration and conceptual discussion purposes only.

Executive Summary: Strategic Transformation

The Vision & Challenge

Dis-Chem is evolving from a traditional retail pharmacy into an integrated primary healthcare ecosystem.

This transition introduces significant operational complexity across pharmaceutical supply chains, requiring real-time, data-driven orchestration.

The Solution & Objectives

The Omni-Channel Inventory Orchestrator is a cloud-native AI platform designed to:

- Optimize working capital allocation across the enterprise.
- Improve inventory efficiency across 300+ stores.
- Reduce pharmaceutical wastage and expiry losses.
- Enable executive-level decision-making through real-time intelligence.

Strategic Context & Business Imperative

Market Transformation

Digital Competition
Rapid entry of digitally native pharmacy players disrupting traditional retail models.

Integrated Healthcare
Growing consumer demand for seamless, patient-centric omni-channel experiences.

Regulatory Complexity
Stringent management requirements for Schedule S1–S6 pharmaceutical assets.

The Risk of Inaction

Failure to modernize inventory systems across Dis-Chem's 300+ stores poses critical threats:

Capital Inefficiency
Excessive tied-up working capital reducing liquidity for strategic reinvestment.

Compliance Exposure
Increased risk of regulatory non-compliance in controlled drug handling.

Competitive Erosion
Loss of market leadership to more agile, data-led healthcare providers.

Financial Impact & Value Creation

Working Capital Optimization

Unlock ~R500 Million

Targeting tied-up inventory capital to significantly improve liquidity and reinvestment capacity across the group.

Inventory Efficiency

10% Reduction

Achieving a reduction in inventory days to increase stock turnover and minimize holding costs at store level.

Margin Preservation

Minimize Write-offs

Reducing losses from expired pharmaceutical products to ensure long-term gross margin sustainability.

Solution Overview: Data + AI Platform

Modern Architecture

Built on a cloud-native Medallion Lakehouse architecture for maximum scalability and governance.

Multi-Source Integration

Seamlessly ingests data from Store POS (JSON), SAP ERP (Parquet), and Clinic systems (CSV).

High-Speed Analytics

Leverages DuckDB and Polars engines for high-performance KPI computation and data processing.

AI Decision Support

LLM-powered agents provide natural language insights and automated inventory recommendations.

Next-Generation Inventory Orchestration

PoC DATA

Enterprise Architecture: Data Flow

Bronze Layer

Raw data ingestion from edge sources without transformation.

Sources:

- Store POS (JSON)
- SAP ERP (Parquet)
- Clinic Systems (CSV)

Silver Layer

Cleaned, standardized, and joined datasets for analysis.

Processes:

- PII Masking (POPIA)
- NAPPI Classification
- Inventory Snapshots

Gold Layer

Business-ready aggregates and high-level metrics.

Outputs:

- Inventory Health Mart
- Financial Performance
- Executive Dashboards

Cloudflare R2 (Lakehouse) | **Cloudflare D1 (Metadata)** | **DuckDB / Polars Compute**

Business Intelligence Framework

Category	Metric	Definition	Data Source
Liquidity	Working Capital Unlocked	Reduction in inventory value vs. baseline period	Gold Layer
Efficiency	Stock Turn Rate	COGS ÷ Average Inventory Value	Gold Layer
Compliance	S6 Dispensing Accuracy	Audit tracking of controlled drug transactions	Silver Layer
Sales	Basket Penetration	Average SKUs per patient visit	Gold Layer

** Framework prioritizes "hard metrics" before introducing AI-driven optimization layers.*

Advanced Analytics & AI Layer

ML Capabilities

Demand Forecasting
Granular SKU-store level predictions using historical sales and behavioral trend modeling.

Inventory Optimization
Dynamic reorder point calculation and safety stock modeling to minimize stockouts.

Seasonal Modeling
Advanced behavioral modeling to account for healthcare seasonality and promotional impacts.

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AI Inventory Agent

Anomaly Detection
Real-time identification of overstock, stockouts, and unusual inventory movements.

Natural Language Insights
Explains complex inventory decisions and provides actionable recommendations in plain English.

Executive Decision Support
Provides real-time, data-driven insights to support high-level strategic planning.

Technology Stack: Efficiency & Scale

Storage & Metadata

Cloudflare R2: Zero-egress storage.
Cloudflare D1: Serverless SQL for metadata.

Compute Engine

DuckDB / Polars: Fast analytical engines for KPI computation.

AI & Intelligence

Workers AI: Llama 3 agents for forecasting and decision support.

Delivery Layer

Streamlit / BI: Interactive dashboards for inventory visibility.

Strategic Cost Advantage

The serverless, edge-native stack delivers up to a **45% reduction** in cloud infrastructure costs versus traditional providers.

PoC - NOT RE

45%

Security, Compliance & Governance

Data Protection

Ensuring full compliance with POPIA and healthcare data regulations through robust masking.

- PII masking at the Silver layer.
- Encryption at rest and in transit.
- Role-based access control (RBAC).

Auditability

Maintaining a transparent record of all automated decisions and system interactions.

- AI decisions logged in Cloudflare D1.
- Immutable audit trails for S1-S6 drugs.
- Real-time monitoring of agent actions.

Traceability

Providing end-to-end visibility from raw data ingestion to executive dashboards.

- Full data lineage tracking.
- Automated metadata management.
- Verifiable KPI computation logic.

ROI & Strategic Conclusion

Target ROI Metrics

Inventory Days	-10%
Cloud Infrastructure Costs	-45%
Pharmacist Productivity	+5 hrs/week

** Targets based on initial architectural projections and benchmarking.*

Strategic Vision

- **Data-Driven Healthcare Leader:** Positioning Dis-Chem at the forefront of the digital pharmacy landscape.
- **Integrated Ecosystem:** Creating a seamless, AI-enabled platform for patient-centric care.
- **Operational Excellence:** Delivering measurable financial impact through advanced orchestration.